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IMAgINE Lab

Adaptive Sequential Feature Grouping and Acquisition for Cost-Efficient Classification

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Introduction — Motivation

Problem

- Features are assumed '*free*' at *inference* — but *costs* are real: financial, time, privacy, energy [1]

Application Domains

- Healthcare: lab tests are costly; clinicians act on informative subsets
- Credit scoring: attributes are unavailable at decision time

Why Group Acquisition?

- Features co-occur naturally in *groups* (lab panels, sensor arrays, questionnaires)
- Acquiring a group jointly yields *higher predictive* value [2]

Instance-Specific Needs

Informative features differ per instance

A *globally fixed* feature subset wastes resources

Goal: Adaptive, cost-efficient strategies to selectively acquire feature groups during inference

Related Work Landscape

Global Feature Selection

Examples: Group Lasso [12], Graph-based [5], VNS (Variable Neighborhood Search) + Markov Blanket [6]

⚠ **Limitation:** Same features for all instances — *limited flexibility*

Instance-wise Feature Selection

Examples: DIWIFT [10], MoE (Mixture of Experts) approaches [9], INVASE [8], COPULA [11]

⚠ **Limitation:** Tailored per instance but *no sequential acquisition* cost model; *selection requires full feature vector upfront*

Group-based Feature Selection

Examples: Group Lasso [12], Elastic Net [13], MoE [14], Learned Encoders (gl) [15]

⚠ **Limitation:** Captures feature dependencies but often *static*

Instance-wise Feature Acquisition

Examples: ETANA [18], Janisch et al. [17], Asgaonkar et al. [19]

⚠ **Limitation:** Sequential but acquires one feature at a time; *lacks adaptive grouping*

SeqFDM addresses all four gaps: instance-wise + group-based + sequential + adaptive stopping

Problem Statement — Setup & Notation

Data Model

Instance $t \in T$: feature vector $V = (V_1, \dots, V_U)^T$, class label $C \in \{C_1, \dots, C_M\}$, class priors $P(C=C_k) = p_k$

Acquisition Cost

$$d_u > 0$$

Cost of acquiring feature V_u

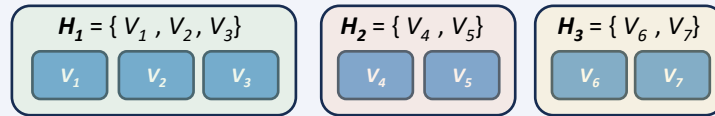
Misclassification Cost

$$N_{kl} \geq 0$$

Penalty for assigning C_l when true class is C_k



Group into super-features



Super-feature (Feature Group) Definition

$$H_q = \{V_{h+1}, V_{h+2}, \dots, V_{h+\delta_q}\}, \quad h = \sum_{f=0}^{q-1} \delta_f, \quad \delta_0 = 0$$

Maximum group size δ is a predefined hyperparameter · Variable size $\delta_q \leq \delta$ allows adaptive grouping per step

Key Assumptions

Fixed ordering

Feature acquisition order is fixed globally across instances

Cond. independence

Features are conditionally independent given class $\rightarrow$ enables tractable Bayesian updates

Max group size δ

Predefined upper bound on super-feature size, keeping solution of associated problem computationally tractable

Problem Statement — Optimization Objective

Decision Variables:

$$\{\delta_q\}_{q=1}^Q$$

group sizes

$$Q$$

stopping group

$$Y_Q \in \{1, \dots, M\}$$

predicted class

Joint Cost Objective — Minimize over $\{\delta_q\}_{q=1}^Q, Q, Y_Q$

$$J\left(\{\delta_q\}_{q=1}^Q, Q, Y_Q\right) = \underbrace{\mathbb{E}\{\Sigma_{q=1}^Q \Sigma_{u=h+1}^{h+\delta_q} d_u\}}_{(I) \text{ feature acquisition cost}} + \underbrace{\Sigma_{l=1}^M \Sigma_{k=1}^M N_{kl} P(Y_Q = l, C = C_k)}_{(II) \text{ expected misclassification cost}}$$

Term (I): Acquisition Cost

Sum of per-feature costs $d_u > 0$ over all acquired groups.
Depends on how many features are acquired and in which groups.

Term (II): Misclassification Cost

$N_{kl} \geq 0$: penalty for assigning C_l when true class is C_k .
Balances classification risk against acquisition savings.

Goal: Find $\{\delta_q\}_{q=1}^Q, Q, Y_Q$ that minimize J — balancing cost of acquiring more features against risk of misclassification

Methodology — Posterior Probability & Bayesian Update

Posterior Vector

After acquiring q super-features: $\zeta_q \triangleq [\zeta_q^1, \dots, \zeta_q^M]^T$, $\zeta_q^k \triangleq P(C = C_k | H_1 = h_1, \dots, H_q = h_q)$

Under conditional independence (A2), likelihood factorizes over features in each group:

$$P(H_q | C_k) = \prod_{u=1}^{\delta_q} P(V_{h+u} | C_k)$$

Recursive Bayesian Update

$$\zeta_q = \frac{\text{diag}(\Phi_q(H_q)) \zeta_{q-1}}{\Phi_q^T(H_q) \zeta_{q-1}}, \quad \zeta_0 = [p_1, \dots, p_M]^T$$

where $\Phi_q(H_q) = [P(H_q | C_1), \dots, P(H_q | C_M)]^T$ and $\text{diag}(\cdot)$ extracts diagonal elements

Methodology — Optimal Classification & Stopping Cost

Given posterior ζ_Q at stopping stage Q , optimal classification minimizes expected misclassification cost:

Optimal Class Decision

$$Y_Q^{opt} = \arg \min_{1 \leq l \leq M} [N_l^T \zeta_Q], \quad N_l \triangleq [N_{1l}, N_{2l}, \dots, N_{Ml}]^T$$

Substituting Y_Q^{opt} , optimization problem simplifies — it now depends only on $\{\delta_q\}_{q=1}^Q$ and Q :

Reduced Objective (over $\{\delta_q\}_{q=1}^Q, Q$ only)

$$\tilde{J}(\{\delta_q\}_{q=1}^Q, Q) = \mathbb{E}\{ \Sigma_{q=1}^Q \Sigma_{u=h+1}^{h+\delta_q} d_u + g(\zeta_Q) \}$$

$$\text{Stopping Cost: } g(\zeta_Q) = \min_{1 \leq l \leq M} [N_l^T \zeta_Q]$$

Key Insight: $g(\zeta_q)$ is expected cost of classifying right now using posterior ζ_q

Methodology — Dynamic Programming

Solve optimal stopping problem over stages $q = \{0, 1, \dots, U\}$ using backward induction:

DP Recursion — $\tilde{J}_q(\zeta_q) = \min$ expected cost from stage q onward

$$\tilde{J}_q(\zeta_q) = \min \left[\underbrace{g(\zeta_q)}_{\text{stop now}}, \underbrace{\tilde{A}_q^1(\zeta_q), \tilde{A}_q^2(\zeta_q), \dots, \tilde{A}_q^z(\zeta_q)}_{\text{continue}} \right], \quad \tilde{J}_U(\zeta_U) = g(\zeta_U)$$

Continuation Cost $\tilde{A}_q^z(\zeta_q)$ — minimized over next group size $z \in \{1, \dots, \delta\}$

$$\tilde{A}_q^z(\zeta_q) = \underbrace{\sum_{u=h+\delta_q+1}^{h+\delta_q+z} d_u}_{\text{cost of acquiring } q^{\text{th}} \text{ super-feature}} + \underbrace{\sum_{H_{q+1}} \Phi_{q+1}^T(H_{q+1}) \zeta_q \tilde{J}_{q+1}(\zeta_{q+1})}_{\text{expected future cost}}$$

Optimal Next Group Size

$$\delta_{q+1} = \arg \min_{z \in \{1, \dots, \delta\}} \tilde{A}_q^z(\zeta_q)$$

Stopping Rule: $Q^{opt} = \text{first } q < U \text{ such that } g(\zeta_q) \leq \tilde{A}_q(\zeta_q)$. If no such q : $Q^{opt} = U$ (acquire all features).

Methodology — SeqFDM Algorithm

Training Phase (Offline)

Step 1: Rank features by descending variance



Step 2: Discretize posterior space with precision η

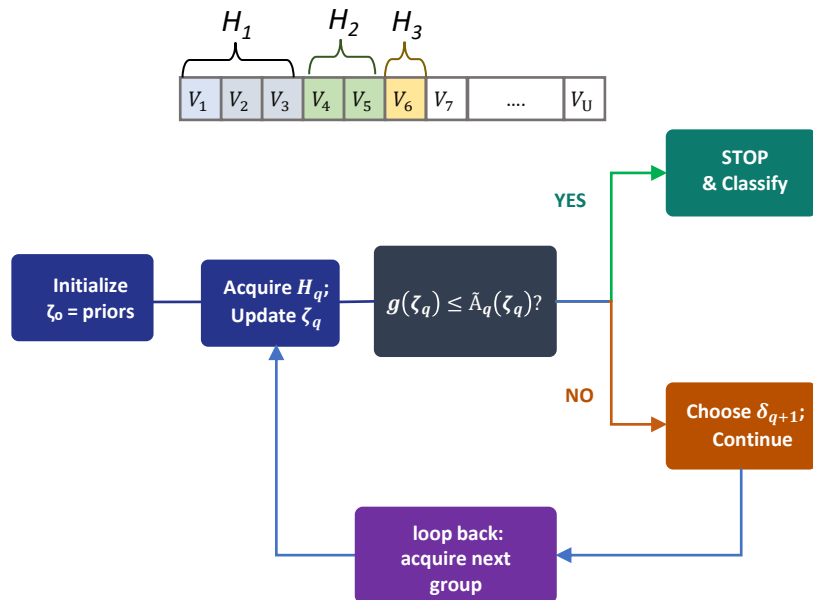


Step 3: Estimate likelihoods $p(V_u|C_k)$ and class priors p_k



Step 4: Solve DP backward — compute $J_q(\zeta_q)$ for all q, ζ_q

Inference Phase (Online)



Experimental Setup

Nine Publicly Available Datasets

Dataset	Instances	Features	Classes
MLL	72	5,848	3
Dexter	300	20,000	2
Leukemia	72	7,129	2
Prostate	102	12,600	2
Lung2	203	12,600	5
Madelon	2,600	500	2
Spambase	4,601	57	2
Eye	14,980	14	2
Diabetes	418	8	2

Baselines

ETANA [18]

Sequential single feature acquisition via DP; 4 bins, $\epsilon = 0.01 \rightarrow$ **instant-wise, sequential**

GLasso [12]

Static group sparse regularization; group size 3, 10^3 iterations $\rightarrow$ **global, group structured**

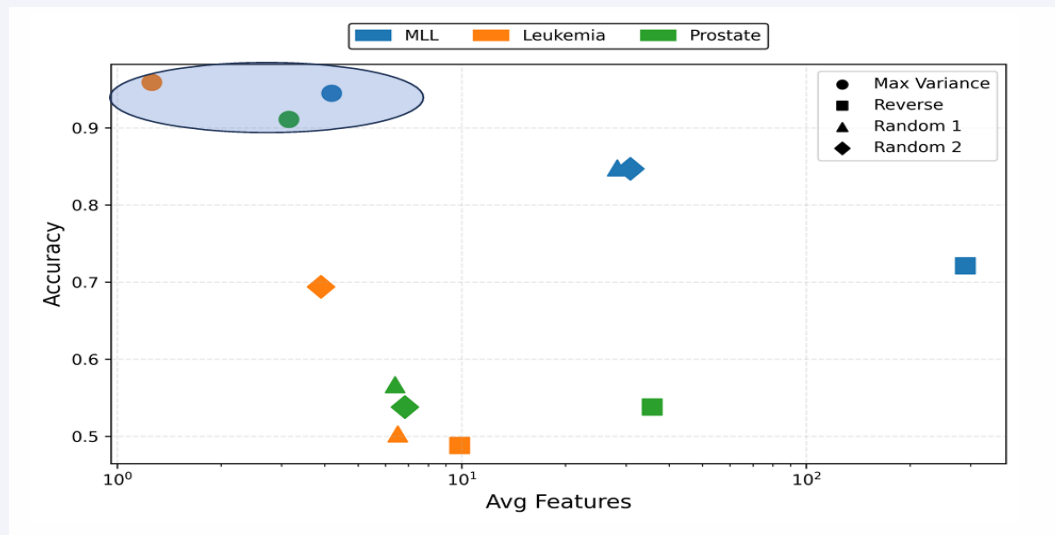
gI [15]

Encoder + variational MI lower bound + Gumbel-Softmax; $k=2$, $m=1 \rightarrow$ **instant-wise, group structured**

Metrics: Accuracy · Avg. # Features · Inference Time | Protocol: 5-fold cross-validation | $N_{kl} = 1 (k \neq l)$, $N_{kk} = 0$

Feature Ordering Strategy

Reduces sequential acquisition from exponential pathfinding → *stopping problem*



Max Variance ✓

Acquires most informative features first

Reverse Ordering

Starts with least informative features

Random Ordering

No structure in acquisition

Key Insight: Fixing global ordering localizes adaptivity to *how many* features to acquire, not *which permutation* to explore — making *DP tractable*

Hyperparameter Analysis

Feature Cost d

$d = 0.15$ provides best accuracy-economy trade-off

Group Size δ

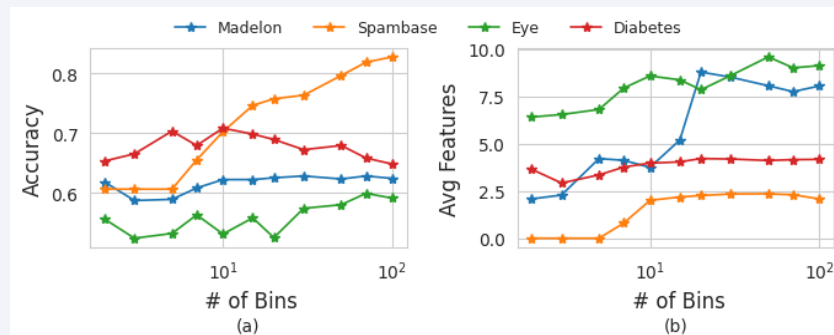
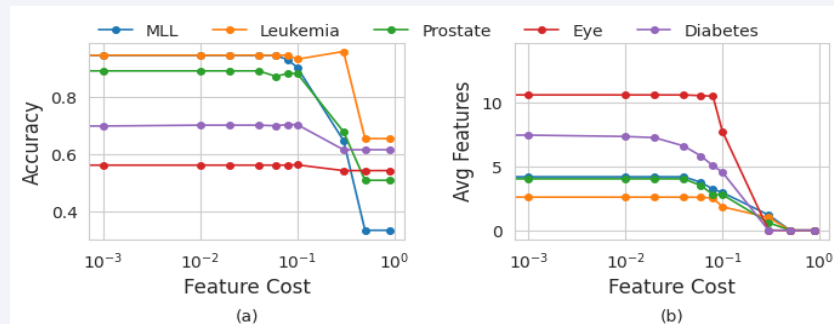
$\delta = 3$ effectively balances acquisition vs. including less informative features

Bin Count V

Generally robust to V

Group Discount

Stable across $[0.0 - 0.3]$.



Default settings used: $d = 0.15$, $\delta = 3$, $V = 4$ bins, discount = 0.1 | Dataset-specific cost adjustments: Spambase ($d=0.008$), MLL ($d=0.001$), Lung2 ($d=0.03$)

Results — SeqFDM vs ETANA

Dataset	Acc SeqFDM	Acc ETANA	# Feat SeqFDM	# Feat ETANA	Time SeqFDM	Time ETANA
MLL	0.945	0.945	4.19	4.21	0.001	0.002
Dexter	0.843	0.767	16.84	17.85	0.010	0.022
Leukemia	0.959	0.916	1.26	2.54	0.001	0.001
Prostate	0.911	0.873	3.15	4.07	0.002	0.002
Lung2	0.842	0.867	17.23	16.45	0.010	0.018
Madelon	0.618	0.616	2.09	3.64	0.018	0.053
Spambase	0.746	0.651	2.18	3.01	0.042	0.077
Eye	0.562	0.560	6.11	0.80	0.230	0.119
Diabetes	0.691	0.694	3.58	3.70	0.005	0.008

Key Findings:

- +4% accuracy on Leukemia & Prostate with 50% / 23% fewer features
- +9% accuracy on Spambase with one-third fewer features
- ~50% faster inference time on most datasets

Results — SeqFDM vs gl

Dataset	Acc SeqFDM	Acc gl	# Feat SeqFDM	# Feat gl	Time SeqFDM	Time gl
MLL	0.945	0.792	4.19	3227	0.001	410.2
Dexter	0.843	—	16.84	—	0.010	—
Leukemia	0.959	0.721	1.26	3511	0.001	576.1
Prostate	0.911	0.810	3.15	6265	0.002	3020.5
Lung2	0.842	0.902	17.23	4148	0.010	5568.1
Madelon	0.618	0.521	2.09	291.2	0.018	17.1
Spambase	0.746	0.661	2.18	25.75	0.042	2.4
Eye	0.562	0.594	6.11	3.50	0.230	3.5
Diabetes	0.691	0.656	3.58	4.21	0.005	0.83

+33%

Accuracy on Leukemia

99.9%

Fewer features vs gl on Leukemia

×2,700+

Faster inference (Prostate: 0.002s vs 3020s)

Memory fail

gl on Dexter (>350 GB RAM)

Results — SeqFDM vs Group Lasso

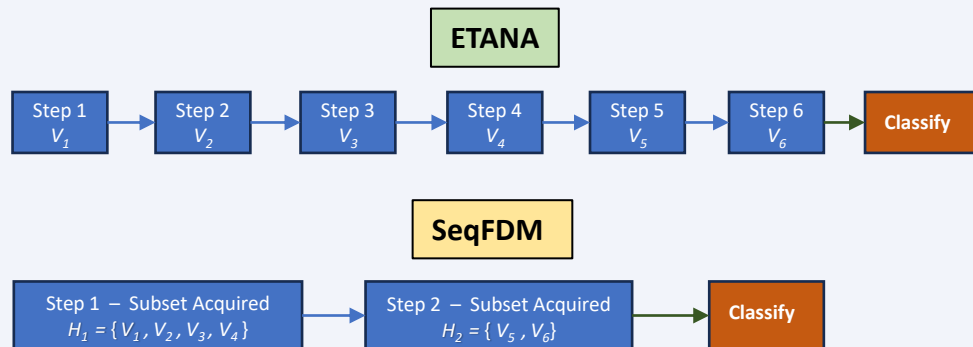
Dataset	Acc SeqFDM	Acc GLasso	# Feat SeqFDM	# Feat GLasso
MLL	0.945	1.000	4.19	7.67
Dexter	0.843	0.920	16.84	239.2
Leukemia	0.959	0.847	1.26	311.8
Prostate	0.911	0.509	3.15	216.6
Lung2	0.842	0.650	17.23	153.8
Madelon	0.618	0.605	2.09	3.40
Spambase	0.746	0.884	2.18	49.82
Eye	0.562	0.629	6.11	14.00
Diabetes	0.691	0.744	3.58	8.00

SeqFDM advantage: On Leukemia: **+11%** accuracy with **<0.5%** of GLasso's features. On Prostate: **+79%** relative gain with **<2%** of features. GLasso is static and instance-agnostic.

Heterogeneous Feature Costs — Disease Prediction Dataset

- Dataset: 42 classes, 132 features each with associated cost; first 3 classes (120 instances each) selected.

Method	Accuracy	# Features	Avg. Cost
SeqFDM	0.983	3.486	0.039
ETANA	0.983	3.237	0.037
gl	0.575	85.74	1.082
GLasso	1.000	8.100	0.163



SeqFDM vs gl

+70.9% accuracy
96.4% cost reduction
27.7× cost factor

SeqFDM vs GLasso

57% fewer features
4.2× cost reduction

SeqFDM vs ETANA

36% fewer acquisition steps
via adaptive grouping

SeqFDM achieves 98.3% accuracy while substantially reducing acquisition cost across all baselines

Instance-wise Acquisition — SeqFDM vs gl (Diabetes Dataset)

DATA INSTANCE A

Predicted label

SeqFDM

Skin Thickness	Pregnancies	Diabetes Pedigree Function	Age	BMI	Blood Pressure	Insulin	Glucose
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No



gl

Skin Thickness	Pregnancies	Diabetes Pedigree Function	Age	BMI	Blood Pressure	Insulin	Glucose
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Yes



DATA INSTANCE B

Predicted label

SeqFDM

Skin Thickness	Pregnancies	Diabetes Pedigree Function	Age	BMI	Blood Pressure	Insulin	Glucose
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No



gl

Skin Thickness	Pregnancies	Diabetes Pedigree Function	Age	BMI	Blood Pressure	Insulin	Glucose
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Yes



Conclusion

Key Contributions

①

SeqFDM Framework

Novel framework for sequential, *instance-wise* acquisition of *variable-sized feature groups* under misclassification–acquisition cost trade-off

②

Optimal Stopping Policy

Derived principled DP-based stopping criterion: stop when $g(\zeta_q) \leq \tilde{A}_q(\zeta_q)$; tractable via fixed ordering

③

Efficient Algorithm

Offline precomputation during training $\rightarrow O(UM + M^2)$ per-instance inference via table lookups; no optimization needed.

④

Empirical Validation

Consistently outperforms ETANA, gl, and GLasso across 9 datasets in accuracy, feature efficiency, and inference speed

Future Directions: Alternative feature orderings · Dependent and Co-related features · Relaxing max group-size constraints

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Thank You!

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Questions & Discussion

Backup Slides — Complexity

Training complexity: $O(U\delta VM\eta^{M-1}) + O(M^2\eta^{M-1})$

Testing complexity: $O(U M + M^2)$ per instance

U : Total number of features

V : # of feature bins

M : # of classes

δ : Max group size

η : Belief space resolution

Backup Slides — Decision Space

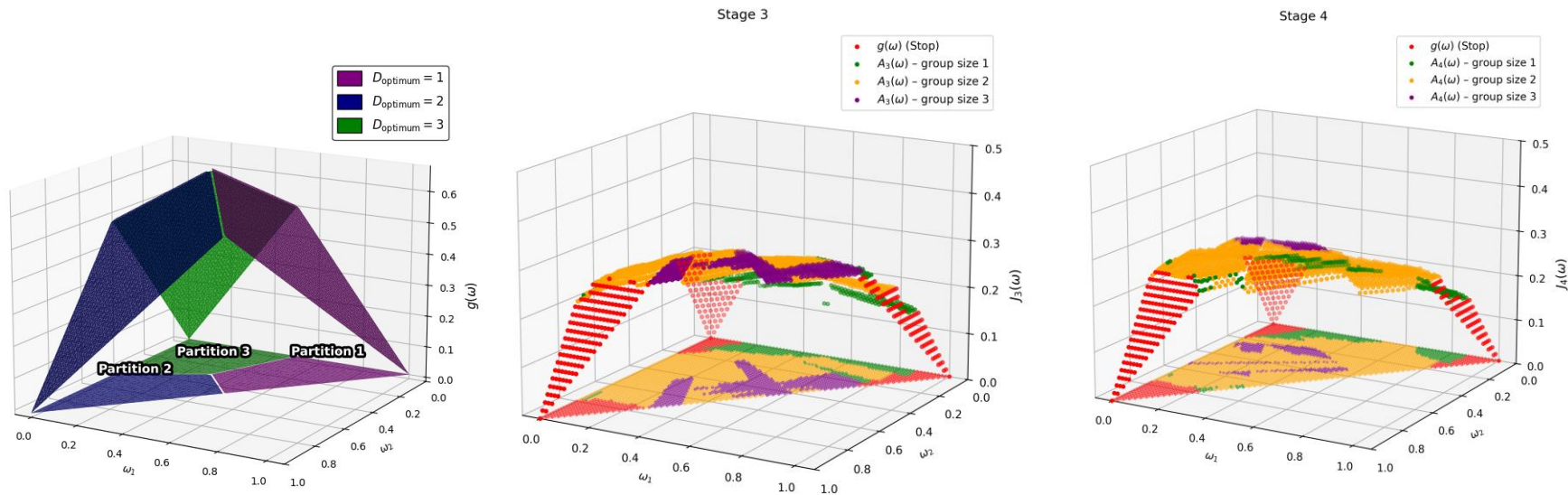


Figure: (a) Probability simplex for 3-class problem and associated decision regions; (b) Decision regions for SeqFDM at Stage 4 and (c) Stage 5, with maximum super-feature size of $\delta = 3$, evaluated on MLL dataset containing 3 classes and 5,848 features. Misclassification costs are set as $N_{kl} = 1$ for $k \neq l$ and $N_{kl} = 0$ for $k = l$.

Backup Slides — Decay Factor

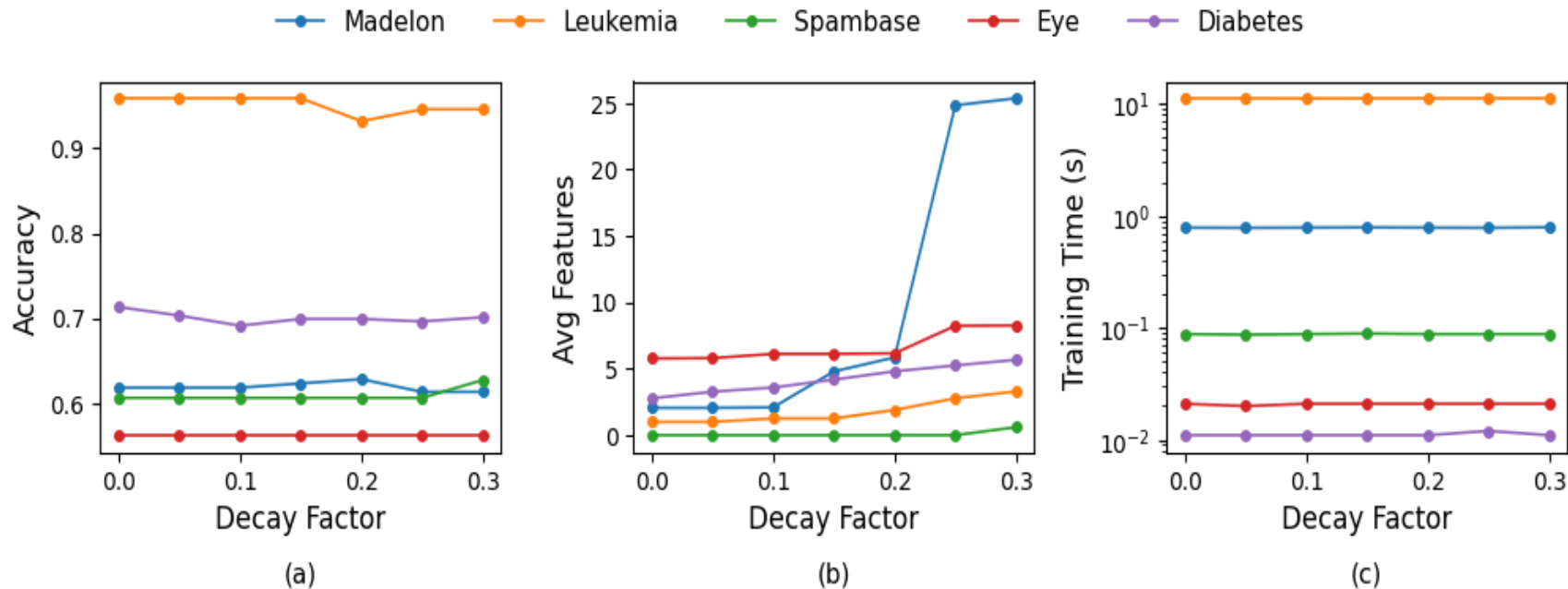


Figure: Impact of decay factor on SeqFDM performance: (a) classification accuracy, (b) average number of acquired features, and (c) training time (s). A decay factor of 0.10 provides best tradeoff.